

Classification Of EEG Signals By ICA And OVR-CSP

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Abstract—Signal processing of Electroencephalography (EEG) plays an important role in brain-computer-interface (BCI) system. Therefore, it is a key in selection of suitable methods. This paper proposed a new method by combining Independent Component Analysis (ICA) with One Versus the Rest Common Spatial Patterns (OVR-CSP) to improve the classification performance. Firstly, EEG signals were filtered with FIR band-pass filter 8-30HZ. Secondly, relative frequency band signals (i.e. μ and β rhythm) were decomposed into independent components to obtain the solution matrix by ICA, and the EEG signals were reconstructed by the main components for improving the signal-to-noise ratio. In order to capture the essential structure of the data, OVR-CSP was used to extract the feature of EEG signals and reduce data dimensions. Finally, using Support Vector Machines (SVM) to classify the feature. The experiment result shows that the average accuracy rate of EEG signals based on motor imagery by the proposed method could achieve 95.555%. It can be understood that ICA is a very effective method to remove artifacts in EEG.

Keywords- EEG; FastICA; OVR-CSP

I. INTRODUCTION

The ultimate goal of brain-computer-interface (BCI) is to translate intentions of a subject into a control signal for a device, say, a computer application, a wheelchair or a neuroprosthesis [1]. It is helpful to these disable people with thought normal but the motor nervous information interruptible. We record brain activity by means of multi-electrode Electroencephalography (EEG) [2-5] which is non-invasive as opposed to invasive work.

Research shows unilateral motor imagery could cause an energy increase of μ rhythm and β rhythm in the ipsilateral hemisphere and lead to an energy decrease at the same time in the contralateral hemisphere, which is known as event related synchronization (ERS) and de-synchronization (ERD) [6-9]. This kind of difference in the sensorimotor area can be applied in EEG analysis. In this paper, we studied EEG signals based on the motor imagery. Because of different motor imagery tasks, EEG patterns are different. However, background noises such as blink artifacts and power noise strongly influence estimation and analysis of EEG signals.

Due to the obvious spatial distribution between signal source and noise source in EEG signals, it is necessary to utilize spatial information within multiple channel signals to preprocess EEG data on the purpose of increasing the signal-to-noise ratio. So far, commonly used methods for spatial filter

are Local Average Technique (LAT), Common Average Reference (CAR), an algorithm for bipolar and the surface Laplacian filter. However, the performance of these methods is not very well.

In this paper, we choose Independent Component Analysis (ICA) for spatial filter. ICA is a signal processing technique that separates independent components based on the principle of statistical independence from multi-channel observable signals in recent years. In biomedical signal processing, EEG signals are mixing with background noises. In other words, EEG signals can be regarded as mixing signals of independent components from different sources. So it is suitable to apply ICA in EEG to obtain the useful signal.

Now, the type of motor imagery in the literature reports more about hands, little about feet and tongue. However, the more the pattern is, the more complex the function of BCI is. Moreover, the utility of BCI is better. In this paper, we contribute by combining FIR filter and ICA, with One Versus the Rest Common Spatial Patterns (OVR-CSP) [10]. We apply our method to classification problem of four class motor imagery EEG signals. This method links two spatial filters ICA and Common Spatial Patterns (CSP), and improves upon the conventional methods in the classification performance on left hand, foot and tongue without much harm to other classification performance.

The rest of paper proceeds as follows. We first provide a short overview over the ICA and OVR-CSP in section II, and then data acquisition and experiment classification are described in section III. Subsequently, experiment results and discussions are shown in section IV. Finally, the description of the conclusion follows in section V.

II. METHODS

A. EEG data preprocessing based on ICA

In order to reserve the characteristics associated with the movement of μ rhythm and β rhythm, EEG data were filtered with FIR band-pass filter 8-30HZ. Fig. 1 (a) and (c) depict the original EEG data and their frequency spectrum of channel C3, C4. Fig. 1 (b) and (d) show the filtered data with FIR filter and their frequency spectrum of channel C3, C4. It was obvious that waveforms of EEG signals in time domain became smooth after FIR filter and EEG signals concentrated in the relative frequency band.

Then we apply ICA to remove noise in this frequency band. ICA is a separation of mixtures of sources with little prior information. The aim of ICA is to seek a linear transformation of these sources to reveal and obtain the hidden

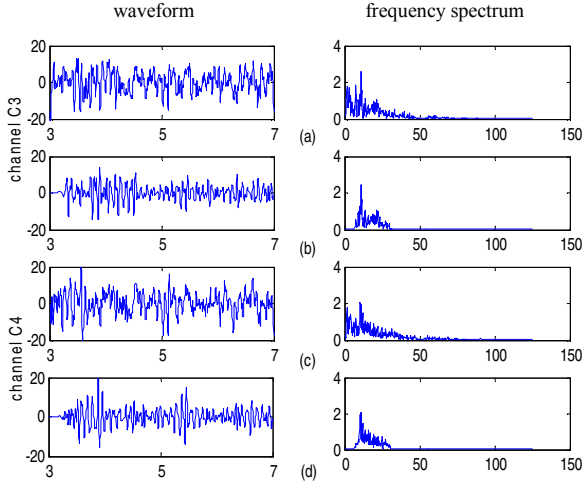


Figure 1. (a), (c) the original EEG signal and its frequency spectrum in channel C3, C4 respectively; (b), (d) the filtered EEG signal with FIR filter and its frequency spectrum in channel C3, C4 respectively.

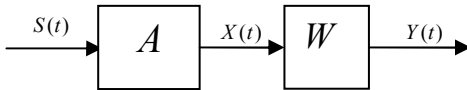


Figure 2. ICA model

factors among them. The model of ICA can be described as Fig. 2.

Where $S(t) = [s_1(t), \dots, s_m(t)]^T$ denotes a series of independent source signals, A denotes mixing matrix, $X(t) = [x_1(t), \dots, x_n(t)]^T$ denotes observed signals, W denotes un-mixing matrix. In the ICA method, the only information we can know is the observed signal $X(t)$. The task of ICA is to find un-mixing matrix W by maximizing some measure of independence to make the followed equation established.

$$Y(t) = WX(t) = WAS(t) \quad (1)$$

We know that under the ideal condition, if $W = A^{-1}$, and $Y(t) = S(t)$. The key of ICA is to establish objective function that measures the independence of separation results and designs optimization algorithm to extract independent sources. Several algorithms have been proposed for this purpose [11], In the paper, the FastICA algorithm based on Negentropy was discussed [12].

EEG signals which come from some different sources were recorded at multiple scalp sensors. So here we need to run FastICA algorithm for several units. One method is to estimate the independent components one by one [13-14].

The steps of FastICA algorithm are followed:

- (1). Make the original signals $X(t) = [x_1(t), \dots, x_n(t)]^T$ to zeros mean, then whiten and get Z .
- (2). Assume m number of independent components to be estimated. Let $p = 1$.
- (3). Take a random initial vector $w_p(0)$, and let $\|w_p(0)\|_2 = 1$.
- (4). Iteration:

$$w_p(k+1) = E\{zg[w_p^T(k)z]\} - E\{g^T[w_p^T(k)z]\}w_p(k).$$
- (5). Gram-Schmidt like de-correlation:

$$w_p(k+1) = w_p(k+1) - \sum_{j=1}^{p-1} \langle w_p(k+1), w_j \rangle w_j.$$
- (6). Let $w_p(k+1) = w_p(k+1) / \|w_p(k+1)\|$.
- (7). If $|w_p^T w_p|$ is not close enough to 1, go back to step (4).
- (8). Let $p = p + 1$, If $p \leq m$, go back to step (3). Otherwise, the task is over.

It is worth noting that the aim of ICA in analysis of EEG signals to find independence components (ICs) with a large proportion corresponding to cortex active regional. The inv of un-mixing matrix (i.e. the column in W^{-1}) can reflect space distribution of these independence sources. So ICs are kept and the remaining are set zeros, when value of space distribution (i.e. the corresponding column in W^{-1}) in the any location within relevant electrode is the first or the second largest in this column. It can be interpreted in Fig. 3 that the IC can be kept when the value of corresponding column is the first or the second largest in the any position C3, C4, Cz (i.e. channel 9, 10, 11) of EEG electrodes and the remaining are kept zeros. Therefore, IC2, IC4, IC6, IC8, IC11, IC14, IC19 are useful signals and can be kept. Finally, we can get EEG signals by FastICA preprocessing.

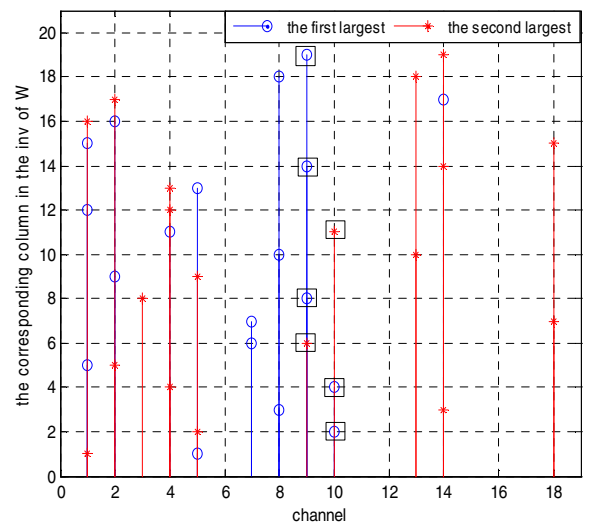


Figure 3. space distribution among the inv of unmixing matrix

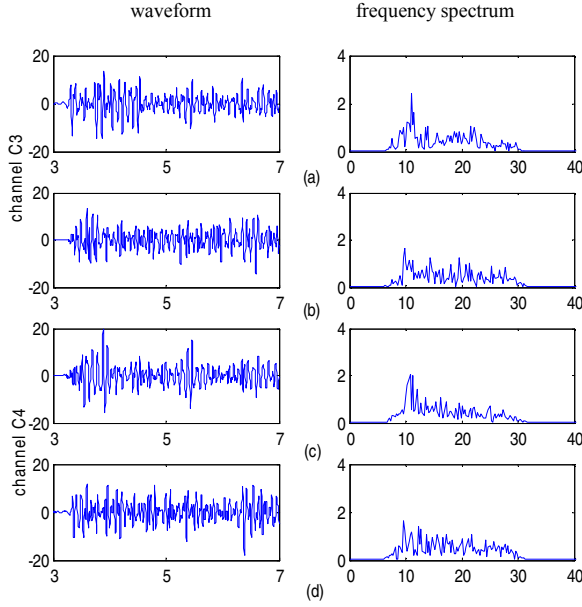


Figure 4. (a), (c) the filtered EEG signal with FIR filter and its frequency spectrum in channel C3, C4 respectively; (b), (d) the spatial filtered EEG signal by ICA and its frequency spectrum in channel C3, C4 respectively.

Fig. 4 shows an example of FIR filtering and ICA preprocessing. As displayed in Fig. 4 (b), (d) the variations of waveform and frequency spectrum by ICA preprocessing were slower than the filtered EEG signal with FIR filter (Fig. 4 (a), (c)), and especially in pass-band 8-30HZ the energy was far stability by contrast Fig. 4 (a), (b) and figure(c), (d). Also relative frequency band did not change.

B. OVR-CSP

OVR-CSP is the extension of CSP. The idea of CSP [15] algorithm is to find the most discriminative spatial filters based on covariance matrix diagonalization simultaneously that maximize variances for one class and minimize variances for the other class at the same time. The limitation for CSP is only designed for two classes. So in this paper, we adopt OVR-CSP for four classes.

We assume that recorded EEG signal matrices can be represented under four different conditions as X_1, X_2, X_3, X_4 with dimension N by K , where N is the number of channels and K is the size of samples in the time series. The estimation of normalized covariance matrix under these four conditions can be given by

$$R_c = \frac{1}{|\varphi_c|} \sum_{i \in \varphi_c} \frac{X_i X_i^T}{\text{trace}(X_i X_i^T)} \quad (c \in \{1, 2, 3, 4\}) \quad (2)$$

Where $\varphi_c (c \in \{1, 2, 3, 4\})$ is the set of trials under each condition and $|\varphi_c|$ denotes the size of a set φ_c .

Then we construct joint covariance matrix

$$R = R_1 + R_2 + R_3 + R_4 \quad (3)$$

The joint covariance matrix can be factored by

$$R = U_0 \Lambda U_0^T \quad (4)$$

Where U_0 denotes a matrix of eigenvectors and Λ represents a diagonal matrix with eigenvalues of R on the main diagonal. Then the whitening matrix can be get by

$$P = \Lambda^{-1/2} U_0 \quad (5)$$

To obtain spatial pattern under condition a , let $R'_a = R_b + R_c + R_d$. So we can get

$$S_a = P R_a P^T \quad (6)$$

$$S'_a = P R'_a P^T \quad (7)$$

It can be proved [16] that S_a and S'_a share the same eigenvector and that the sum of corresponding eigenvalues for the two matrices is always one. It can be understood that if S_a can be factored as

$$S_a = U \Lambda_a U^T \quad (8)$$

Then

$$S'_a = U \Lambda'_a U^T \quad (9)$$

And

$$\Lambda_a + \Lambda'_a = I \quad (10)$$

Combine (6) (7) (8) and (9), then obtain

$$\Lambda_a = (P^T U)^T R_a (P^T U) = S F_a R_a S F_a^T \quad (11)$$

$$\Lambda'_a = (P^T U)^T R'_a (P^T U) = S F_a R'_a S F_a^T \quad (12)$$

Where $S F_a = U^T P$. Note that (10), it is obvious that larger eigenvectors corresponding to larger eigenvalues yield a high variance under condition a and a low variance under other conditions.

With the projection matrix $W_a = U^T P$ we can get

$$Z_a = W_a X_a \quad (13)$$

Repeat the above procedure, we can obtain Z_b, Z_c, Z_d

C. Classification

Finally, the features used for classification are obtained by choosing both ends of the eigenvector matrix U . In practice, only a small part of these eigenvectors are selected. Assume $U^* = (U_1, \dots, U_m, U_{N-m+1}, \dots, U_N)$ and $m \ll N$. Then we get signals $Z_p = (U^{*T} P) X$, and these signals are m first and last rows of Z . Subsequently, we compute feature vectors to be sent to classifier in (14).

$$f_p = \log \left(\frac{\text{var}(Z_p)}{\sum_{i=1}^{2m} \text{var}(Z_i)} \right) \quad (14)$$

At last, Support Vector Machines (SVM) [17-19] is used as classifier model.

III. DATA ACQUISITION AND CLASSIFICATION OF EXPERIMENT

Our method has been used for the EEG data from BCI competition 2005 data IIIa. The data was provided by the Laboratory Clinical of Brain-Computer Interfaces (BCI-Lab), Graz University of Austria. The recording concluded 60

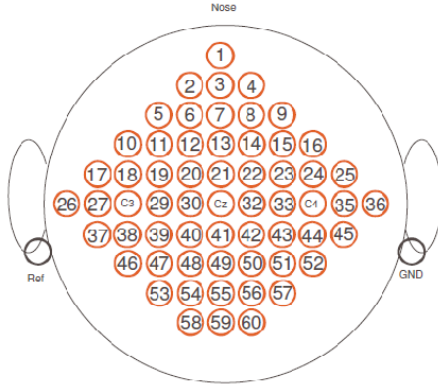


Figure 5. Position of EEG electrodes.

channels and was sampled with 250HZ. The task was to perform imagery four movements (left hand, right hand, foot, tongue) according to a cue from three persons (k3b, k6b, l1b). The data sets consist of 360 trials, 240 trials and 240 trials respectively. The experiment included three periods: the first 2s were quite; at 2s, cue appeared; from 3s to 7s, the subject began to imagine action.

In order to save memory space and reduce computation time, we selected 19 channels(12、14、17、19、21、23、25、26、28、31、34、36、37、39、41、43、45、48、50) [20] from 60 channels. Position of EEG electrodes is shown in Fig. 5. The data are chosen from 3s to 7s. In OVR-CSP, the common spatial filter is build with two eigenvectors from both ends of eigenvector matrix, and these spatial filters can be formed by distinction between one class and the remaining other classes. Therefore, the number of common spatial filters to be obtained is three. Then the feature vectors can be obtained in terms of (14) after spatial filter. Finally, we choose three SVM to classify these features.

In this paper, since the thinking state is different among different persons, we only choose person k3b for our experiment.

IV. RESULTS AND DISCUSSION

In this paper, we use classification accuracy rate of test data to quantify the feasibility and performance of the method. TABLE I. shows the experiment result and the result only by FIR filter preprocessing is also listed in order to contrast here.

From TABLE I, it can be seen that our proposed method can improve the classification accuracy rate effectively, especially in the left hand by 9% and tongue by approximately 4% and by approximately 2% in tongue respectively. The average accuracy rate can increase by 3% and the results can reveal that ICA has a good performance in the multiple classes EEG signal preprocessing. In Fig. 4, we can see that ICA can

effectively remove artifacts according to contrast in time domain and frequency domain.

In order to exact the feature, we apply OVR-CSP. At last, SVM is used as classifier. It is worth mentioning to that ICA has three disadvantages: first, ICA can not determine the variances of independence components because of assuming unit variance; second, the order of the independence

TABLE I. THE CLASSIFICATION ACCURACY RATE OF DIFFERENT TYPES PREPROCESSING

methods	Left (%)	Right (%)	Foot (%)	Tongue (%)	Average accuracy rate (%)
FIR+ICA	93.33	97.78	97.78	93.33	95.555
FIR	84.44	100	93.33	91.11	92.22

components can not be determined; third, it has uncertainty of the amplitude.

V. CONCLUSIONS

The method in this paper can enhance the classification performance by preprocessing the EEG data with a linear spatial filter ICA. EEG signals for multiple channels consist of independence sources and volume conduction among these channels also exists. This makes ICA successfully separates background noises and useful signals, then increases Signal Noise Ratio. Therefore, ICA has many potential in artifacts removed of EEG signals and ICA may become a powerful tool in EEG signals enhancement. The experiment may also prove that the OVR-CSP method is an effective method of feature extraction.

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